

EmpiRE-Compass: A Neuro-Symbolic Dashboard for Sustainable and Dynamic Knowledge Exploration, Synthesis, and Reuse

Amirreza Alasti¹, Lena John¹, Sushant Aggarwal², Yücel Celik², and Oliver Karras¹

¹ TIB – Leibniz Information Centre for Science and Technology, Germany ² Leibniz University Hannover, Germany Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Ujjwal Karn](#)

Reviewers:

- [@theRizwan](#)
- [@prashantcraju](#)

Submitted: 12 April 2026

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

EmpiRE-Compass is an open web application for exploring curated scholarly knowledge about empirical research in software and requirements engineering. It combines structured data in a **research knowledge graph** with optional **large language model** assistance so that users can ask questions, inspect the underlying queries, and view reproducible charts—without treating the model as a source of facts. The software is openly licensed, publicly deployed, and archived with a persistent identifier ([Alasti et al., 2026](#)).

Software engineering (SE) and requirements engineering (RE) face a significant increase in secondary studies, particularly literature reviews (LRs), due to the ever-growing number of scientific publications ([Auer et al., 2023](#); [Karras, 2025](#); [Mendes et al., 2020](#); [Napoleão & others, 2022](#)). The rise of generative artificial intelligence (GenAI) has made ensuring the quality and transparency of LRs increasingly challenging. In general, data analyzed by GenAI systems are often not fully trustworthy or reliable due to issues such as hallucinations ([Schryen et al., 2025](#)).

EmpiRE-Compass is a neuro-symbolic dashboard designed to lower barriers for accessing, replicating, and reusing LR data. By semantically structuring underlying data in research knowledge graphs (RKGs) and leveraging large language models (LLMs) for dynamic access, it enables sustainable knowledge exploration, synthesis, and reuse ([Karras, 2025](#)). The tool builds upon FAIR data principles ([Stocker & others, 2022](#); [Wilkinson & others, 2016](#)) and open scholarly knowledge graph practice ([Auer et al., 2023](#)) to make research data Findable, Accessible, Interoperable, and Reusable.

Statement of Need

The core problem in modern LRs is the lack of availability of underlying data and artifacts ([Karras & others, 2023](#)). This prevents collaboration and updating of LRs, as analyses and results cannot be replicated ([dos Santos & others, 2024](#); [Karras, 2025](#)). While open science practices are growing, only a minority of studies verifiably share their data and artifacts in persistent repositories ([Huotala & others, 2025](#); [Wernlein, 2022](#)), often limited to static files rather than machine-actionable data ([Kaplan & others, 2025](#); [Spadaro et al., 2022](#)). Furthermore, effectively using Research Knowledge Graphs (RKGs) like the **Open Research Knowledge Graph (ORKG)** requires advanced technical skills to query and interpret structured data.

Target audience. EmpiRE-Compass is intended for **researchers, educators, and reviewers** who run or rely on secondary studies (including systematic or living literature reviews) in

41 empirical software engineering and requirements engineering, and who need **traceable** access
42 to community-curated evidence without becoming SPARQL experts.

43 State of the Field

44 Existing tools for scientific knowledge exploration and evidence synthesis generally fall into
45 three categories, each with limitations that EmpiRE-Compass addresses.

- 46 1. **Static and domain-specific dashboards.** Scholarly knowledge-graph dashboards, such as
47 those built on ORKG for specific domains, demonstrate that KG-powered visualizations
48 can improve usability and engagement for targeted topics, but they typically expose a
49 fixed set of charts and filters for a single template or dataset, rather than enabling users
50 to formulate new questions or compose multi-template syntheses across studies (Auer et
51 al., 2023). Similarly, “living” SLR dashboards in domains like biomedicine show the value
52 of continuously updated, interactive evidence summaries, yet they are usually tightly
53 coupled to bespoke pipelines and do not expose a reusable KG-centric representation or
54 user-editable query logic (Elbers et al., 2021; Manion et al., 2024).
- 55 2. **Neuro-symbolic scholarly assistants.** Recent neuro-symbolic systems for scholarly search,
56 including those that combine ORKG with large language models, focus primarily on
57 answering natural language questions or ranking relevant contributions at the document
58 level (Oelen et al., 2024). While these approaches reduce hallucinations by grounding in
59 a KG, they rarely provide transparent, user-steerable analytics workflows: users cannot
60 readily inspect or modify the underlying queries and data-processing steps, and the
61 outputs are typically textual answers rather than reproducible, shareable statistical
62 visualizations over curated LR datasets (Colelough & Regli, 2025; Oelen et al., 2024).
- 63 3. **Generic LLM assistants and graph frontends.** General-purpose LLM tools (e.g.,
64 conversational assistants) can summarize and compare individual papers but lack
65 a symbolic ground truth and frequently hallucinate, making them unsuitable as a
66 basis for rigorous secondary studies, especially when traceability and reproducibility
67 are required (Li et al., 2025). Conversely, generic KG frontends such as the ORKG
68 comparison UI enable users to manually assemble comparisons and tables across papers,
69 but constructing complex, multi-paper syntheses demands substantial expertise in the
70 underlying data model and query concepts and does not scale well as datasets and
71 research questions grow (Auer et al., 2023).

72 **EmpiRE-Compass** positions itself as a neuro-symbolic orchestration layer on top of ORKG that
73 bridges these gaps: it retains the KG as the single source of truth, like existing dashboards
74 and neuro-symbolic systems, but exposes the full pipeline from natural language question
75 to SPARQL query to JavaScript data processing and visualization. This design enables
76 non-technical users to perform complex, multi-paper analytics through natural language, while
77 still allowing experts to inspect, edit, rerun, and share the exact queries and computations that
78 underpin each “living” literature review.

79 Software Design

80 **EmpiRE-Compass** is a Progressive Web Application (PWA) built with a modular architecture.
81 It separates the **Symbolic Layer** (the truth source) from the **Neural Layer** (the interface). The
82 system is available online at <https://empire-compass.tib.eu> and the source code is hosted on
83 [GitHub](#).

84 Architecture & Design Trade-offs

85 We explain the main trade-offs, the design we chose, and why it matters for a research application
86 that combines knowledge graphs with LLMs—balancing automation with correctness, privacy
87 with convenience, and generality with domain fidelity.

Symbolic vs. neural. We treat the **symbolic layer** (ORKG, SPARQL, triplestore) as the **only source of factual data**. The LLM generates SPARQL and interprets only the *returned* bindings; it never invents answers. That choice demands iterative refinement and editable code rather than a hidden pipeline. For literature review and empirical research practice, **grounding in the KG** is essential: conclusions must be traceable to the graph, and our design ensures every finding can be traced back to a SPARQL query and ORKG data.

Human-in-the-loop. We prioritize **accuracy and user control** over full automation. The system generates SPARQL, executes it against the live ORKG endpoint, and evaluates the outcome (syntax errors, empty results, all-null columns). On failure, it feeds structured feedback (including domain-specific hints) back into the LLM and retries up to **three times**, using the **database as the oracle**. Users can **edit the generated SPARQL** and the **data-processing code** (JavaScript) directly in the UI. Making the pipeline visible and editable supports **trust and reproducibility**: the same question can be re-run, shared, or adapted.

Template-driven domain fidelity. We use a **template registry** and **template-specific guidance** per ORKG template (e.g. R186491): domain knowledge, query patterns, and common mistakes are injected into prompts. This adds maintenance when adding templates but improves correctness and keeps the system **extensible**—new themes (e.g. NLP4RE) can be added without changing core logic—and aligns generated queries with how the community models each domain in the KG.

Privacy and key management. A **unified AI service** chooses at runtime: if the user supplies their own API keys, calls go **directly from the frontend** to the provider (OpenAI, Groq, Mistral, Google); if they use environment keys, calls go through our **Node.js backend**. We thus support both **privacy-first** use (sensitive questions never touch our servers) and **shared/demo** use without per-user key setup.

Technology. The frontend uses **React** and **Material-UI**; the backend uses **Node.js** and the **Vercel AI SDK** for model-agnostic LLM calls across providers. **Firestore** holds template metadata and community questions with real-time sync and Keycloak integration. These choices keep the symbolic layer (KG + SPARQL) as the definitive source of truth while supporting maintainability and community-driven use.

Research Impact Statement

EmpiRE-Compass is developed in direct connection to peer-reviewed work on **KG-EmpIRE** (Karras & others, 2023) and public ORKG infrastructure (Auer et al., 2023). It has been applied to two curated research datasets that illustrate end-to-end use for the software engineering community:

1. **KG-EmpIRE**: interactive visualization and analytics over empirical RE literature encoded in the graph (776 papers in the deployment described in the repository documentation).
2. **NLP4RE ID Card**: dynamic access to 49 detailed “ID cards” for NLP-for-RE tools, aligned with replication-oriented discussion in the field (Abualhaja & others, 2024).

The software is **publicly hosted**, versioned, and **archived on Zenodo** with a DOI for citation and reproducibility (Alasti et al., 2026), alongside a machine-readable **CITATION.cff** file in the repository. Together with Storybook-hosted UI documentation and open issue tracking, these artifacts provide concrete, checkable signals of community-oriented release practice beyond a one-off prototype.

By making these datasets interactive, the dashboard supports “living” literature reviews that can be updated as new contributions appear in the underlying knowledge graph, in line with calls for sustainable, evidence-based secondary research (Mendes et al., 2020; Mendez & others, 2020).

AI Usage Disclosure

Manuscript. The authors used **Microsoft Copilot** and **Grammarly** for initial drafting, grammar refinement, and spelling checks of this manuscript. The core software design, architectural decisions, and the neuro-symbolic integration logic were conceived and implemented by the human authors. All AI-generated text was reviewed, edited, and validated by the authors to ensure accuracy and compliance with JOSS standards.

Software. EmpiRE-Compass integrates **third-party large language model APIs** as an optional runtime feature (user-configured providers and keys). During development, team members may have used **AI-assisted coding or refactoring tools**; all changes were reviewed, tested, and validated by the human authors, and the symbolic layer (ORKG/SPARQL) remains the source of factual answers.

Acknowledgements

We thank the Open Research Knowledge Graph (ORKG) team for their infrastructure support.

References

- Abualhaija, S., & others. (2024). Replication in Requirements Engineering: The NLP for RE Case. *ACM Transactions on Software Engineering and Methodology*, 33(6). <https://doi.org/10.1145/3658669>
- Alasti, A., John, L., Aggarwal, S., Celik, Y., & Karras, O. (2026). *EmpiRE-Compass*. Zenodo. <https://doi.org/10.5281/zenodo.18170203>
- Auer, S., Stocker, M., Karras, O., Oelen, A., D'Souza, J., & Lorenz, A.-L. (2023). Organizing scholarly knowledge in the open research knowledge graph: An open-science platform for FAIR scholarly knowledge. *Proceedings of the Conference on Research Data Infrastructure*, 1. <https://doi.org/10.52825/cordi.v1i.272>
- Colelough, B. C., & Regli, W. (2025). *Neuro-symbolic AI in 2024: A systematic review*. <https://arxiv.org/abs/2501.05435>
- dos Santos, V., & others. (2024). Sustainable Systematic Literature Reviews. *Information and Software Technology*, 176. <https://doi.org/10.1016/j.infsof.2024.107551>
- Elbers, S., Wittink, H., Kaiser, U., Kleijnen, J., Pool, J., Köke, A., & Smeets, R. (2021). Living systematic reviews in rehabilitation science can improve evidence-based healthcare. *Systematic Reviews*, 10(1), 309. <https://doi.org/10.1186/s13643-021-01857-5>
- Huotala, A., & others. (2025). Research Artifacts in Secondary Studies: A Systematic Mapping in Software Engineering. *Information and Software Technology*, 187. <https://doi.org/10.1016/j.infsof.2025.107830>
- Kaplan, A., & others. (2025). Structuring Scientific Knowledge in Software Engineering Using the Open Research Knowledge Graph: A Use Case and Experience Report. *Software Engineering 2025 – Companion Proceedings*. <https://doi.org/10.18420/se2025-ws-29>
- Karras, O. (2025). Research Knowledge Graphs for Sustainable Literature Reviews in Software Engineering and Beyond. *Software Engineering 2025 – Companion Proceedings*. <https://doi.org/10.18420/se2025-ws-30>
- Karras, O., & others. (2023). Divide and Conquer the EmpiRE: A Community-Maintainable Knowledge Graph of Empirical Research in Requirements Engineering. *17th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement (ESEM)*. <https://doi.org/10.1109/ESEM56168.2023.10304795>

- Li, Z., Yi, W., & Chen, J. (2025). *Beyond accuracy: Rethinking hallucination and regulatory response in generative AI*. <https://arxiv.org/abs/2509.13345>
- Manion, F. J., Du, J., Wang, D., He, L., Lin, B., Wang, J., Wang, S., Eckels, D., Cervenka, J., Fiduccia, P. C., Cossrow, N., & Yao, L. (2024). Accelerating evidence synthesis in observational studies: Development of a living natural language processing-assisted intelligent systematic literature review system. *JMIR Medical Informatics*, 12, e54653. <https://doi.org/10.2196/54653>
- Mendes, E., Wohlin, C., Felizardo, K., & Kalinowski, M. (2020). When to Update Systematic Literature Reviews in Software Engineering. *Journal of Systems and Software*, 167, 110607. <https://doi.org/10.1016/j.jss.2020.110607>
- Mendez, D., & others. (2020). Open Science in Software Engineering. In *Contemporary empirical methods in software engineering*. Springer. https://doi.org/10.1007/978-3-030-32489-6_17
- Napoleão, B. M., & others. (2022). Towards Continuous Systematic Literature Review in Software Engineering. *48th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*. <https://doi.org/10.1109/SEAA56994.2022.00078>
- Oelen, A., Jaradeh, M. Y., & Auer, S. (2024). *ORKG ASK: A neuro-symbolic scholarly search and exploration system*. <https://arxiv.org/abs/2412.04977>
- Schryen, G., Marrone, M., & Yang, J. (2025). Exploring the scope of generative AI in literature review development. *Electronic Markets*, 35. <https://doi.org/10.1007/s12525-025-00754-2>
- Spadaro, G., Tiddi, I., Columbus, S., Jin, S., ten Teije, A., & Balliet, D. (2022). The Cooperation Databank: Machine-Readable Science Accelerates Research Synthesis. *Perspectives on Psychological Science*, 17(5), 1472–1489. <https://doi.org/10.1177/17456916211053319>
- Stocker, M., & others. (2022). SKG4EOSC - Scholarly Knowledge Graphs for EOSC: Establishing a Backbone of Knowledge Graphs for FAIR Scholarly Information in EOSC. *Research Ideas and Outcomes*, 8, e83789. <https://doi.org/10.3897/rio.8.e83789>
- Wernlein, F. (2022). *Acquisition and Analysis of Research Practices Using Semantic Structures* [Bachelor Thesis], Gottfried Wilhelm Leibniz Universität. <https://doi.org/10.15488/12796>
- Wilkinson, M. D., & others. (2016). The FAIR Guiding Principles for Scientific Data Management and Stewardship. *Scientific Data*, 3(1). <https://doi.org/10.1038/sdata.2016.18>